AI-Powered Robo-Advisor Platform

Academic Project Report
Programming in Finance II — USI, 2026

P1 (Backend/Data) P2 (Quant/Optimizer)
P3 (ML/Profiling) P4 (Frontend/LLM/Docs)

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1 Introduction

Most commercial robo-advisors share three structural limitations: opaque models, undisclosed training data, and unconstrained AI components. This project addresses all three in an academic prototype combining a Gradient Boosting risk profiler trained on the Federal Reserve Survey of Consumer Finances (SCF 2022), a Hierarchical Risk Parity (HRP) portfolio optimizer with Ledoit–Wolf covariance shrinkage, and a constrained LLM narrator (large language model API) that is mathematically prevented from inventing figures via a Ground Truth JSON contract and a five-step validator. An EU Awareness layer (Rule 9) explicitly communicates the US-centric bias of the SCF training data to European investors at the point of recommendation.

The platform is a Streamlit web application backed by a FastAPI service, with a SQLite audit trail storing SHA-256 hashes of every market data snapshot and LLM prompt. An agentic GitHub Actions workflow demonstrates the mandatory AI agent criterion: a pull request generated entirely by the LLM API without human intervention.

Source code and Architecture Decision Records: `https://github.com/Programming-for-finance-II/robo-advisor`

2 Backend and Data Infrastructure (P1)

2.1 API Layer

The backend is a **FastAPI** service exposing five endpoints: `/profile` (risk profiler), `/optimize` (portfolio optimiser), `/advice` (LLM narrator), `/backtest` (historical scenario analysis), and `/compare` (HRP vs. MVO side-by-side). All protected endpoints require an `X-API-Key` header validated on every request. Rate limiting is enforced via `slowapi` at 10 requests per minute per IP address; requests exceeding the limit receive a `429 Too Many Requests` response without consuming any computation. All responses are Pydantic-validated, ensuring schema consistency between the backend and the Streamlit frontend.

2.2 Data Pipeline: ValidatedDataLoader

Market data is downloaded from `yfinance` by the **ValidatedDataLoader** (`backend/data/loader.py`). The loader applies three sequential quality controls before returning data to the optimiser:

1. **UCITS probe.** For each primary UCITS ticker, a lightweight download is performed and the per-ticker NaN ratio is computed. If the ratio exceeds the 2 % threshold, the loader substitutes the corresponding US-listed fallback (e.g. `SPY` for `CSPX.L`) and records the substitution.
2. **Forward-fill.** Gaps of up to two consecutive trading days are filled; remaining NaN values cause a `DataQualityError`.
3. **Minimum observations gate.** At least 252 valid trading days (one calendar year) must remain after cleaning; shorter series are rejected.

The cleaned price matrix is serialised to CSV and hashed (SHA-256) to produce the `market_data_hash` stored in the audit trail.

2.3 Audit Trail

The SQLite database (`robo_advisor.db`) contains two tables. `market_data_snapshots` stores one row per unique `market_data_hash`, using `INSERT OR IGNORE` for deduplication: identical market data downloaded on different days produces a single stored snapshot. `recommendations` stores one row per optimisation request, linking to the corresponding snapshot via the hash and recording the full input parameters, profile output, portfolio weights, and risk metrics. This design enables bit-for-bit reproducibility of any historical recommendation from the stored hash alone. The application is containerised via `docker-compose.yml`; on Streamlit Community Cloud the database resets on redeploy, as documented in ADR-005.

3 ML Risk Profiler (P3)

3.1 Architecture

The profiler operates in two phases. **Phase A** is a rule-based fallback using the Grable–Lytton (1999) instrument (Grable and Lytton 1999): ten questionnaire items scored on $[0, 30]$ are mapped to three profiles (`CONSERVATIVE`, `MODERATE`, `AGGRESSIVE`), with a capital-preservation override that caps the profile regardless of total score. **Phase B** uses a trained GBM classifier grounded in empirically observed household behaviour. Both phases return a uniform `ProfilerOutput` typed dictionary, so downstream components are agnostic to which phase produced the classification.

3.2 Label Construction and GBM Classifier

Training labels are derived from the SCF 2022 (Board of Governors of the Federal Reserve System 2022) ($n = 4{,}595$ households, implicate 1). Four allocation variables (`EQUITY`, `BOND`, `CASHLI`, `STOCKS`) are normalised and clustered via K-Means ($K = 3$, silhouette $s = 0.84$). Clusters are labelled deterministically by mean equity share. A `HistGradientBoostingClassifier` is trained on nine questionnaire-derived features with SCF sample weights preserved. Performance:

Metric	**HistGBM**	**LR Baseline**
Training accuracy	97.7%	79.9%
CV accuracy (3-fold)	94.0% ± 0.15%	63.3% ± 2.9%

Table 1: Classifier performance on SCF 2022.

3.3 SHAP Explainability

SHAP values are computed via `TreeExplainer` and normalised by absolute magnitude. The top-N drivers are injected into the Ground Truth JSON; the LLM narrator is constrained to reference only features present in `top_drivers` (Rule 7), preventing fabricated correlations.

4 Portfolio Optimisation (P2)

4.1 ETF Universe

The platform allocates across eight ETFs providing multi-asset diversification under MiFID II constraints, with three UCITS-compliant primary tickers and US-listed fallbacks for the remainder.

Table 2: ETF universe — primary tickers, clusters, UCITS status.

Ticker	Asset class	Cluster	UCITS
CSPX.L	Equity USA	risk_assets	✓
EFA	Equity International	risk_assets	
GLD	Gold	real_assets	
VNQ	REIT USA	real_assets	
AGGH.MI	Bond Aggregate (EUR)	safe_haven	✓
TLT	US Treasury 20Y+	safe_haven	
TIP	TIPS Inflation-Linked	safe_haven	
XEON.MI	Cash / Overnight	cash	✓

4.2 Hierarchical Risk Parity

HRP (López de Prado 2016) was selected over MVO for three structural reasons (ADR-001): it requires no matrix inversion, no expected-return estimate, and produces no corner solutions. The algorithm proceeds via Ward-linkage tree clustering on a Ledoit–Wolf distance matrix, quasi-diagonalisation, and recursive bisection assigning weights inversely proportional to intra-cluster variance:

$$\alpha_L = 1 - \frac{V_L}{V_L + V_R}, \qquad V_k = \mathbf{w}_k^\top \hat{\Sigma}_k \, \mathbf{w}_k$$

Raw weights are tilted by risk profile: Conservative blends toward Minimum Variance; Aggressive toward Equal Risk Contribution (Maillard, Roncalli, and Teïletche 2010); Balanced uses raw HRP weights.

4.3 Ledoit–Wolf Shrinkage and Constraints

The Oracle Approximating Shrinkage estimator (Ledoit and Wolf 2004) $\hat{\Sigma} = (1-\alpha)S + \alpha\mu_S I$ is applied before clustering to stabilise the distance matrix. Box constraints clip weights to $[5\%, 40\%]$ per asset and $[10\%, 60\%]$ per cluster; weights are renormalised after clipping.

4.4 Market Regime Detection

A dual-trigger detector flags `HIGH_STRESS` when $\bar{\rho}_{\text{LW}} > 0.75$ or VIX > 30 (ADR-006). Under stress, HRP is replaced by a cluster-level Equal Risk Contribution portfolio and a banner is shown in the frontend.

5 LLM Narrator and Validator (P4)

5.1 Ground Truth JSON Contract

All numerical computations are performed by the backend. The LLM receives a `GroundTruthPayload` containing pre-computed figures and is permitted only to narrate them. The payload includes portfolio weights, risk metrics (volatility, max drawdown, VaR, CVaR), profiler output with SHAP drivers, an `allowed_numbers` whitelist (auto-populated by recursively extracting every float from the payload), and regulatory context. `expected_annual_return` and `sharpe_ratio` are intentionally `null` for HRP, as the algorithm produces no reliable forward-return estimate.

5.2 System Prompt: Nine Absolute Rules

The narrator operates under nine rules enforced at prompt level: (1) no invented numbers outside `allowed_numbers`; (2) no prescriptive advice (forbidden phrases injected from the payload); (3) no compliance claims or absolutes; (4) historical framing only for return figures; (5) fixed out-of-context response; (6) cluster descriptions in economic, not correlation, terms; (7) SHAP drivers only; (8) mandatory disclaimer; (9) **EU Awareness** — when `profiler_us_centric_caveat` is true, the narrator must acknowledge SCF US-centrism and European investor differences.

5.3 Five-Step Validator

Every response passes through `validator.py` before reaching the user. Steps 1, 2, 4, 5 are blocking; Step 3 is corrective:

1. **Forbidden phrase check** — case-insensitive substring scan.
2. **Hallucinated number check** — all floats extracted via regex, percentages normalised, checked against `allowed_numbers` within $\pm 2\%$ tolerance.
3. **Disclaimer presence** — auto-appended if absent.
4. **Injection detection** — semantic scan of output for known injection patterns.
5. **EU Awareness (Rule 9)** — verifies SCF/Federal Reserve reference and European investor reference are both present when required.

5.4 Prompt Injection Defence

Layer 1 sanitises user input before the API call: inputs exceeding 500 characters are rejected; known injection keywords are blocked case-insensitively. Accepted input is wrapped in `<user_input>` tags. Layer 2 is Validator Step 4, catching cases where the model echoes injected instructions in its output.

6 Backtest Results

Three historical stress scenarios are evaluated using US proxy tickers with 10 bps transaction costs per rebalance.

Table 3: Backtest summary — HRP vs. MV vs. $1/N$ (MODERATE profile).

Scenario	Model	Ann. Return	Volatility	Sharpe	Max DD
GFC (2008)	HRP	−0.1%	8.6%	−0.01	−11.9%
	MV	−5.7%	10.7%	−0.53	−18.0%
	$1/N$	−5.1%	18.4%	−0.28	−25.1%
COVID-19 (2020)	HRP	+6.4%	7.7%	0.83	−10.1%
	MV	+7.2%	13.8%	0.52	−15.5%
	$1/N$	+11.9%	13.3%	0.90	−19.1%
Rate Hike (2022)	HRP	−8.9%	7.6%	−1.17	−13.0%
	MV	−13.1%	10.4%	−1.27	−16.3%
	$1/N$	−13.7%	11.3%	−1.21	−19.1%

HRP consistently delivers lower volatility and smaller drawdowns than both benchmarks across all three scenarios, at the cost of a modest return penalty relative to $1/N$ in the COVID recovery.

7 Limitations

yfinance fragility. The platform relies on an unofficial API with no SLA. UCITS-listed tickers occasionally return excessive NaN values; the `ValidatedDataLoader` mitigates this with automatic fallback, but live operation remains dependent on third-party data availability.

SCF US-centrism. The risk profiler is trained on US household data. European investors may exhibit materially different risk preferences, savings rates, and financial literacy patterns. The EU Awareness layer communicates this explicitly at the point of recommendation, but does not correct the underlying distributional mismatch.

HRP opacity. The recursive bisection algorithm produces weights that cannot be summarised by a simple rule. The LLM narrator and the MVO comparison tab partially mitigate this, but residual opacity is a fundamental limitation acknowledged in ADR-001.

Validator false positives. The substring check for *safe* also matches the cluster label *safe_haven*, triggering unnecessary fallback responses. Whole-word matching is deferred to future work. The EU Awareness check (Step 5) is keyword-based rather than semantic and may fail on paraphrased compliant responses.

SQLite persistence. The audit trail resets on Streamlit Cloud redeploy. A managed PostgreSQL instance would be required in production (ADR-005).

8 Conclusions

This paper presented an educational robo-advisor prototype demonstrating a complete investment workflow — from risk profiling to portfolio optimisation to natural-language explanation — while explicitly addressing transparency and geographic-awareness gaps common in commercial platforms. The Ground Truth JSON contract and the five-step validator represent the core architectural innovation: they make hallucination prevention structural rather than instructional. The EU Awareness layer

(Rule 9, `profiler_us_centric_caveat`, Validator Step 5) is a deliberate design choice to surface the geographic limitations of a US-trained model to European investors at the point of recommendation.

Future Work

Replace the SCF 2022 training dataset with the ECB Household Finance and Consumption Survey (HFCS); replace `yfinance` with a reliable data provider; implement whole-word matching in the validator; add narrator retry logic before returning the safe fallback; extend the regime detector with a rolling confirmation window.

References

Board of Governors of the Federal Reserve System (2022). *Survey of Consumer Finances, 2022*. Accessed May 2026. URL: https://www.federalreserve.gov/econres/scfindex.htm.

European Parliament and Council (2014). *Directive 2014/65/EU on Markets in Financial Instruments (MiFID II)*. URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32014L0065.

Grable, John E. and Ruth H. Lytton (1999). "Financial Risk Tolerance Revisited: The Development of a Risk Assessment Instrument". In: *Financial Services Review* 8.3, pp. 163–181.

Ledoit, Olivier and Michael Wolf (2004). "A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices". In: *Journal of Multivariate Analysis* 88.2, pp. 365–411.

López de Prado, Marcos (2016). "Building Diversified Portfolios that Outperform Out-of-Sample". In: *Journal of Portfolio Management* 42.4, pp. 59–69. DOI: 10.3905/jpm.2016.42.4.059.

Maillard, Sébastien, Thierry Roncalli, and Jérôme Teïletche (2010). "The Properties of Equally Weighted Risk Contribution Portfolios". In: *Journal of Portfolio Management* 36.4, pp. 60–70.